

Phase 2 – Requirement Analysis

AI WeatherWise Project

1. Software Requirements

- Windows 10/11, macOS or Linux.
- Node.js v16 or above.
- npm v8 or above.
- Gemini AI SDK / documented Gemini integration.
- MongoDB and Mongoose.
- Postman for API verification.
- Visual Studio Code or similar IDE.

2. Hardware Requirements

- Intel Core i5 (8th Gen+) / AMD Ryzen 5 or equivalent.
- 8 GB RAM minimum; 16 GB recommended.
- 1 GB available storage.

3. Functional Requirements

- User registration and login.
- JWT authentication and protected routes.
- CRUD operations for favorite locations.
- Fetch current weather metrics.
- Request AI weather summaries and recommendations.
- Admin user-account management and monitoring.

4. Quality / Security Requirements

- bcrypt password hashing.
- JWT validation and role-based authorization.
- Centralized error handling.
- Request/payload sanitization.
- Fallback mode when external API keys are unavailable.
- Modular and maintainable backend structure.

5. Technology Stack

Node.js + Express.js + MongoDB + Mongoose + JWT + bcrypt + Gemini AI + Postman/Thunder Client.